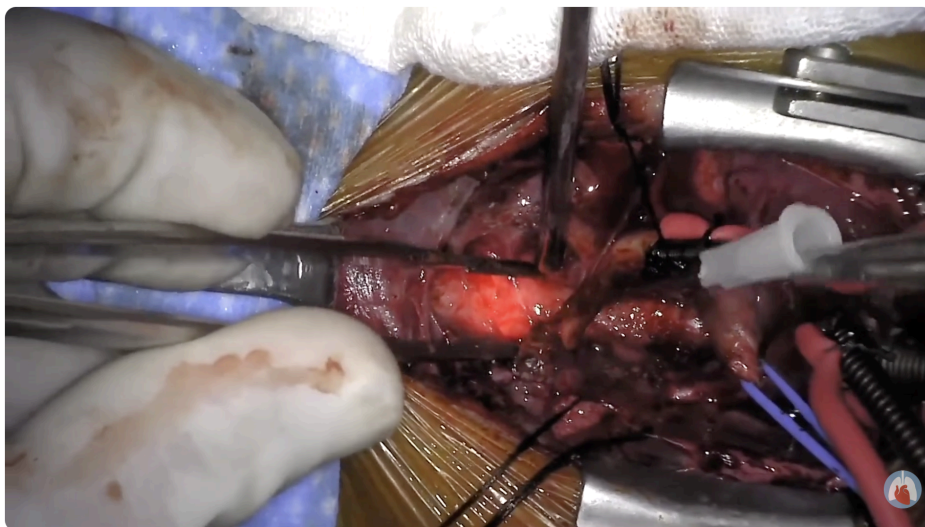
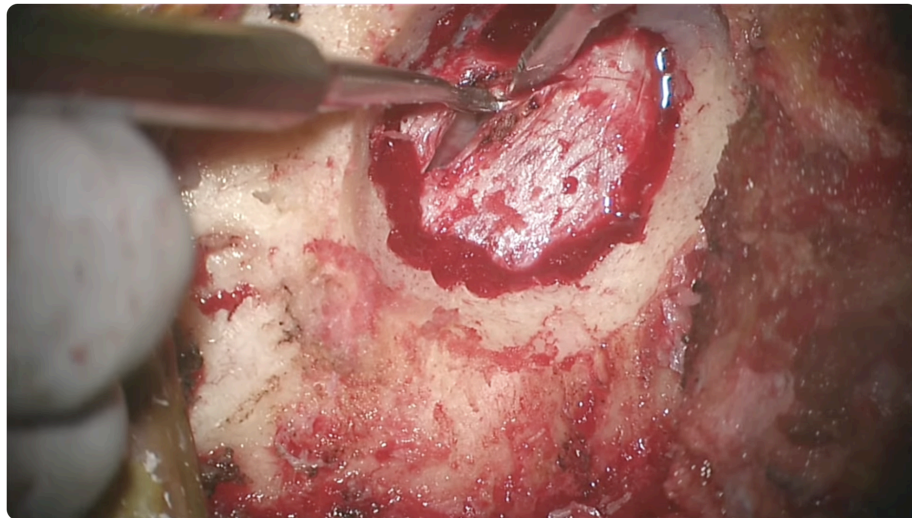


Key Features to Distinguish between Surgical Techniques:

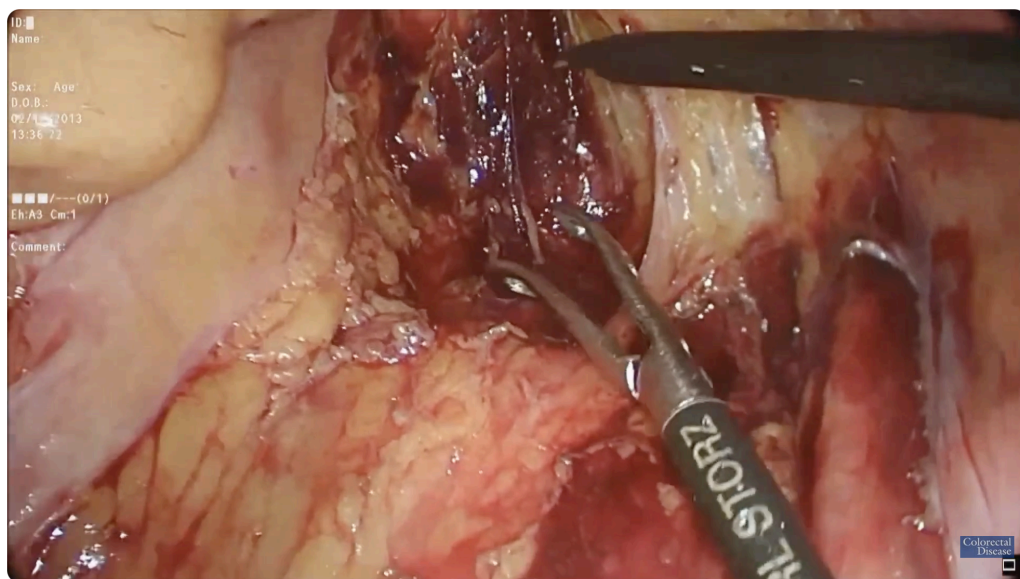
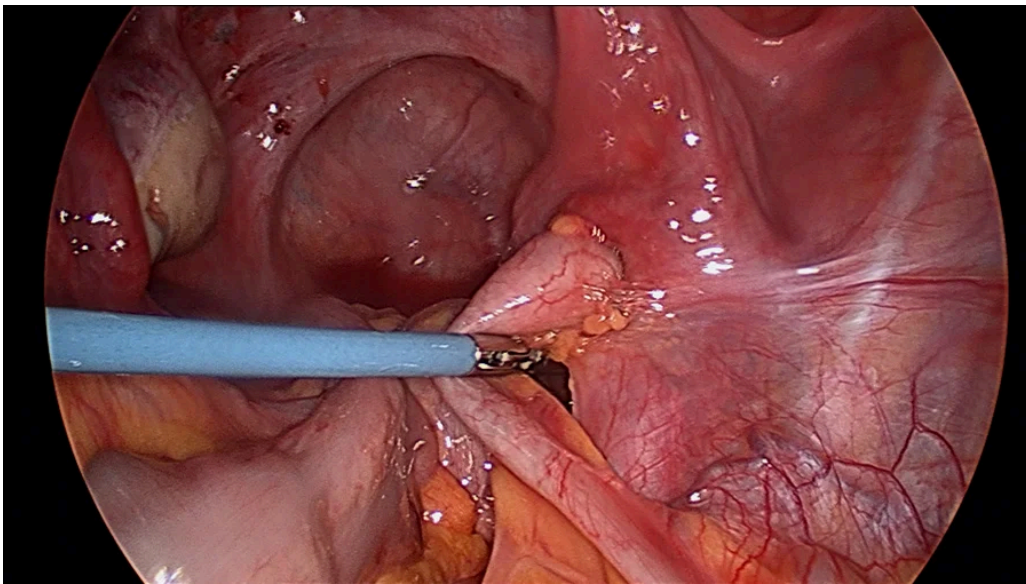
Open Surgery:

- Are gloves visible?
 - If yes, open surgery
- Large incision
- Tissue held open with retractors
- Blood and anatomy fully exposed
- No circular camera border
- Often microscope or exoscope view in neurosurgery
- Gauze/sponges visible
- Surgeon's fingers manipulating tissue
- Wide surgical field



Laparoscopic:

- Usually circular camera view (not always, so make sure to double check)
- Long thin instruments entering from edges
 - Often for laparoscopic surgeries, there is AT MOST 2 tools being used, excluding the camera (again, not always, but usually)
- Camera moves inside abdomen
 - Sometimes it is shaky because the attending surgeon is manipulating the view; the movement of the camera is not “angular”
- **No hands visible inside body**
- Ports/trocars



Robotic:

- Looks similar to laparoscopic, which makes it harder to differentiate, but usually square view
- **Extremely stable camera**
 - The zoom, repositioning is very very efficient, almost little to no shaking
- Instruments have **jointed/wristed movement**; can be more than 2 instruments inside at one given time
- Words that suggest ROBOTIC:
 - rTAPP
 - Robot-assisted
 - da Vinci
- Look for specific tools, the **Bipolar Forceps** are the easiest tool to spot and distinguish; picture attached

